

Abstract No.

Development of a multimodal platform for the characterization of glioblastoma circulating tumor cells using FLIM Analysis

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Background: Glioblastoma multiforme (GBM) is an aggressive malignancy where liquid biopsy, specifically circulating tumor cell (CTC) isolation, offers a promising non-invasive tool for longitudinal disease monitoring. However, capturing GBM CTCs remains challenging due to their rarity and lack of epithelial markers. This study develops a workflow combining microfluidic enrichment, nanostructured slide immobilization, and Fluorescence Lifetime Imaging Microscopy (FLIM) to phenotypically and metabolically characterize GBM CTCs.

Methods: The platform was initially validated using a spike-in model of T98G GBM cells mixed with healthy lymphocytes. The cell suspension was deposited onto Tethis slides prior to immunofluorescence (IF) staining using Cell-Surface Vimentin (CSV) as the tumor marker and CD45 for lymphocytes. Crucially, this pipeline has already been successfully applied to clinical samples: whole blood from GBM patients was enriched via the Parsortix system, immobilized on Tethis slides, and stained for CSV and CD45. Extracted cells were quantified using an Olympus confocal microscope and analyzed via autofluorescence FLIM for metabolic profiling and DAPI-FLIM for chromatin conformation.

Results: Spike-in experiments demonstrated efficient cell adhesion on the Tethis platform. IF staining successfully differentiated T98G tumor cells from the surrounding CD45+ background. Real patient sample experiments using the Parsortix system confirmed the effectiveness of the protocol for CTC isolation and identification. Preliminary FLIM analyses revealed a unique metabolic signature in tumor cells compared to healthy lymphocytes, while DAPI-FLIM optimization showed distinct variations in chromatin architecture between the two populations.

Conclusions: This integrated pipeline successfully couples morphological, phenotypic, and metabolic single-cell analysis. Validated in both spiked models and initial clinical samples, the platform represents a powerful tool for characterizing GBM CTCs and tracking disease progression.



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